Figure 16.10 A convex function with a single, and proper, global minimum.

Why is that? Well, you can imagine an optimization algorithm like a marble that is left rolling on the surface of the cost function; there are, of course, marbles of different weight and friction, and some marbles on some surfaces get stuck and need to be pushed around – likewise, there are different learning algorithms.

Anyway, if you release a marble on a surface looking like the cost function in figure 16.9, there is a good chance it just stays on the plateau where it lands; if you give it a little push, it might end up in a pit corresponding to a local minimum, and be stuck there, unless you pick it up and land it elsewhere.

Conversely, if you release a marble on a smooth surface looking like figure 16.10, like when you toss it into a bowl, then it will roll down to the bottom of the bowl, maybe oscillate a little, and in the end settle down in the lowest point, where gravity can't pull it down any further.

16.2.3 Optimizing Random Sampling

Using this "marble" analogy, an optimization problem can be seen as two things:

- The cost function has an analogous in the marble track: the smoothest its path to the optimal cost, the better algorithms can (in theory) work. Notice that we can build several "tracks" between a starting point and a finish line: engineering the best possible track is part of the solution.
- An optimization algorithm is like a marble rolling down the path: for the analogy to be more accurate, though, we need to say that both the marble, and the way that's tossed and